

Course Syllabus

General Information

Faculty	Information Technology		
Department	Computer Science		
Semester	Summer	Academic Year	2024/2025

Course Details

Course Title	Object-Oriented Programming in Java			Course No.	0412201		
Type of Learning	☐ On- campus ☒ Blended (..... Platform)			☐ Online (..... Platform)			
Credit Hours	3	Theoretical	3	Practical	0		
Course Level (according to JNQF)			6	JNQF Hours*		12	
Pre-requisite	Object-Oriented Programming C++- 0411203			Co-requisite			

Course Instructor Information

Instructor Name	Dr. Samer Odeh Hanna		
Instructor E-mail	shanna@meu.edu.jo		
Course Time	09:30 – 11:00 Sunday to Wednesday		
Office Hours	(11:00-12:00 Sunday, Monday)		
Office Number	N143	Office Phone	Ext. 311
Lab Instructor Name –If assigned-			

Sources and References

1. Textbook

Absolute Java, Walter Savitch, Pearson, 6th edition, 2016

2. References

- “Java Programming: From Problem Analysis to Program Design”, D.S.Malek, 5th edition, 2012
- “Java How to Program”, Deitel H. M. and Deitel P. J., 7th edition, prentice hall, 2007.
- “Introduction to Java Programming and Data Structures”, 12th edition. Y Daniel Liang by Pearson 20

Teaching Methods

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|---------------------------|------------------------------|--------------------------------|
| 1. Lecture-based learning | 2. Technology-based learning | 3. Individual / Group learning |
| 4. Problem-based learning | 5. Projects/Presentations | 6. Discussions |

Course Description

This course introduces basic Object oriented concepts and techniques. The fundamental concepts and techniques encapsulation, inheritance, polymorphism, interfaces, and abstract classes are the foundation for object oriented programming. Building the foundation prepares students to learn object-oriented programming and advanced Java programming.

Course Learning Outcomes (CLOs) and its correlation with Program Learning Outcomes (PLOs) and JNQF Descriptors

No.	CLOs	PLOs	JNQF Descriptors		
			Knowledge	Skill	Competence
1	Describe the main concepts of the object-oriented programming paradigm as they are related to software development.	1	✓		
2	Use Classes, Control Statements, Functions, Recursion and Arrays in a program that incorporates these concepts.	1,2		✓	
3	Write programs that incorporate object oriented constructs and features such as classes and objects, abstraction, encapsulation, inheritance and polymorphism.	1		✓	
4	Illustrate the concept of object oriented programming in solve problem and how to write the required program code to solve it.	1,2			✓
5	Apply Object oriented programming concept in problem solving.	1,2			✓
6	Manipulate Exception Handling & String.	1,2			✓

Course Topics

Week	Topic	Lecture Type		CLOs	Assessment	
		Lecture No.	Lecture Type		Assessment Tool	Summative Formativ
1	Course Syllabus Discussion & Introduction to Java	Lecture (1)	L	1	-	F
		Lecture (2)	P	1	-	F
2	JAVA (Basic elements, Identifiers Variables, Assignment Statements)	Lecture (1)	L	1	1,3	S
		Lecture (2)	P	2	1,3	S
3	JAVA (Program Style, Loops, Control statements)	Lecture (1)	L	1	1,3	S
		Lecture (2)	P	2	1,3	S
4	JAVA (Console Input and Output, Flow of Control, Array, Function)	Lecture (1)	L	1	1,3	S
		Lecture (2)	P	2	1,3	S
5	Arrays JAVA (Classes 1, Class Definitions)	Lecture (1)	L	1	1,3,4	S
		Lecture (2)	P	3	1,3,4	S
6	Object Oriented Programming (Methods, Overloading, Constructors)	Lecture (1)	L	1	1,3,4	S
		Lecture (2)	P	3	1,3,4	S
7	Object Oriented Programming- Classes 2 (Encapsulation, this keyword, Static methods and variables inheritance) Object Oriented Programming (inheritance)	Lecture (1)	L	3	1,3,4	
		Lecture (2)	P	4	1,3,4	S
Week	Topic	Lecture Type		CLOs	Assessment	
		Lecture No.	Lecture Type		Assessment Tool	Summative Formativ
8	Object Oriented Programming (inheritance) Mid-Term	Lecture (1)	L	2	2,3,4	S
		Lecture (2)	P	4	2,3,4	S
9	Object Oriented Programming (inheritance)	Lecture (1)	L	1, 3, 4	2,3,4	S
		Lecture (2)	P	1, 3, 4	2,3,4	S
10	Object Oriented Programming (polymorphism)	Lecture (1)	L	2, 3, 4,5	2,3,4	S
		Lecture (2)	P	3, 4,5	2,3,4	S
11	Abstract Classes and Interfaces (Chapter 8)	Lecture (1)	L	2, 3, 4,5	2, 4	S
		Lecture (2)	P	2, 3, 4,5	2, 4	S
12	Exception Handling & String Chapter 9	Lecture (1)	L	2, 3, 4,5	2, 4	S
		Lecture (2)	P	2, 3, 4,5	2, 4	S
13	Files I/O & Multithreading Chapter 11 & 12	Lecture (1)	L	6	2, 4	S
		Lecture (2)	P	6	2, 4	S
14	GUI Basics: Introduction to JavaFX or Swing (Chapter 12)	Lecture (1)	L	6	2, 4	S
		Lecture (2)	P	6	2, 4	F
15	Advanced Topics in OOP & Project Development Chapter 14	Lecture (1)	L	5	4	F
		Lecture (2)	P	4,5		

Course Evaluation Time and Weight

Assessment Tool	Weight %	Expected Due Date
1 Mid Written Exam.	25	W8

2 Final Written Exam.	40	W16
3 Assignments & Quizzes	25	W3,W5,W7,W11,W13
4 Project.	10	W14

Course Policies

Item	Policy
Quizzes	- No makeup quizzes. - A minimum of 2 quizzes are given. - Each Quiz is out of 5. - If 3 quizzes or more are given then the lowest quiz's grade is dropped. - Pop quizzes may be administered periodically throughout the semester.
Exams	The format of the exams is generally (but NOT always) as follows: General Definitions, Multiple-Choice, True/False, Analyze a Problem, Essay Questions, etc.
Makeup Exams	Makeup exam should not be given unless there is a valid excuse.
Drop Date	Last day to drop the course is according to the University regulations.
Academic Honesty	- Cheating or copying on exam or quiz is an illegal and unethical activity. - Standard MEU policy will be applied. - All graded assignments must be your own work (your own words).
Attendance	- Excellent attendance is expected. - MEU policy requires the faculty member to assign ZERO grade if a student misses 10% of the classes. - Sign-in sheets will be circulated. - If you miss class, it is your responsibility to find out about any announcements or assignments you may have missed.
E-Learning	It is your responsibility to check the course's eLearning website regularly (at least once every 2-3 days) for announcements and material.
Workload	The student should expect to spend 6 hours per week as an average workload.
Graded Exams	Instructor should return exam papers graded to students within one week after the exam date.
Participation	- Participation in and contribution to class discussions will affect your final grade positively. Raise your hand if you have any question. - Making any kind of disruption and (side talks) in the class will affect you negative
Assignment/ Projects	- Homework's are as mentioned in the previous table. - Students organize themselves in teams of (2-3) students each. Each team will give at least one presentation regarding a selected case, delivering a fully working application. - More details about phases and topics of the projects will be given later on.
Others	

Rubrics and Feedbacks (If Applicable)

* JNQF Hours Calculation

Activity	Duration/ week	Times/ Semester	No. of Hours
Learning Activities			

Lectures and Seminars	3	14	42
Tutorials	4	2	8
laboratory	–	–	–
Activities for Assessment and Evaluation			
First Exam/ Midterm	1.5	1	1.5
Second Exam (If applicable)	–	–	–
Final Exam	2	1	2
Quizzes	1	2	2
Self-Learning Activities			
Assignments/ Homework	3	3	9
Case Studies	3.5	1	3.5
Presentations	1	2	2
E-learning	1.5	14	21
Extra Readings	3	10	30
Total Hours	121		
JNQF Hours	=Total Hours/10 =12.1		